



# Increased prevalence of Parkinson's disease in soils with high arsenic levels

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## ABSTRACT

**Introduction:** Exposure to arsenic (As)-containing pesticides was associated with an increased risk for Parkinson's disease (PD). Arsenic also induced in murine brains  $\alpha$ -synuclein aggregates, a pathognomic feature of PD.

**Objectives:** People living on farms irrigated with high As water in Taiwan are likely exposed to increased As. We addressed whether increased farm soil As levels correlate with an increased PD risk.

**Methods:** We used the information from several national surveys (1983–1986) on soil metal contents to study the relationships between soil metal contents and PD prevalence in Taiwan. PD prevalence (2009–2013) was calculated with a database from Taiwan's compulsory national health insurance. A patient is defined by a PD diagnosis and prescriptions of Levodopa in three or more office visits in twelve months. We used regression models to study the correlation between PD prevalence and soil metal contents.

**Results:** The PD prevalence ranged from 83 to 213 per one hundred thousand persons in different regions of Taiwan. Among the studied heavy metals, we found only As was significantly associated with the PD prevalence. The top three regions (Yunlin, Chiayi, Tainan) in the PD prevalence list correspond exactly with the top three in soil As levels. Soil As levels and PD prevalence had a strong correlation ( $r = 0.75$ ).

**Conclusion:** PD prevalence is statistically correlated with farm soil As levels in Taiwan.

## 1. Introduction

Parkinson's disease (PD) is one of the most prevalent neurodegenerative diseases in the world [1]. Despite decades of research, its cause remain elusive. Increased incidence was found in those frequently exposed to As-containing pesticides [2]. Meanwhile, results from animal studies support a probable involvement of As in PD. Aggregates of synuclein in the brain were elevated in mice with As exposure [3].

There are many arsenic-containing pesticides, e.g., methylarsonic acid and ferric methylarsonate. Although they were banned in Taiwan in 2018. They are widely used in growing rice and cotton in the world [4, 5]. Despite past use of these arsenic-containing chemicals in agriculture, they compose only a minor part of the exposure source of arsenic. Geologic studies found that there is a big underground lake in the southwest part of Taiwan. The lake water contains high concentrations of arsenic. In the past, people did not have the opportunity to drink the lake water or use it for irrigation, because it is lying hundreds of meters under sea level [6]. Residents on the southwest coast of Taiwan where water was scarce, began to dig wells as deep as 200 m in the 1920s after the technology became available. The As concentrations in water from

these wells (more than 2000) ranged from 0.35 to 1.14 ppm [7], which are much higher than the standard limit for arsenic in drinking water by the US Environmental Protection Agency (0.01 ppm) [8].

Researchers in Taiwan have made great efforts to study the effects of As intake via drinking the well water on human health. Arsenic was first linked with Blackfoot disease [9], and later the links were expanded to cancer, cardiovascular and metabolic diseases [10–12]. The arsenic research has lead to the revision of international health standards for arsenic exposure.

Farm soils contain high As levels in some parts of the region as a result of decades long irrigation with As-contaminated water. People living on these contaminated soils are highly likely to be exposed to increased levels of As via farm products. One study showed that relative risks for certain types of cancer in the region remained high, despite the government's having constructed a municipal water system for the region in the 1970s [13].

Heavy metals have been linked with PD [14]. Despite some studies showed a positive link between certain metal and the disease, others showed results against a link. Incessant publication of papers on this issue indicates its importance and demands more studies to clarify the

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pathogenic roles of heavy metals in PD. Here we addressed whether increased farm soil As levels correlated with an increased PD risk.

## 2. Methods

### 2.1. Study population and definition of PD patients

In 1995 the National Health Insurance Research Database (NHIRD) was established by the National Health Insurance (NHI), a system of comprehensive medical services for Taiwan residents. The NHIRD, maintained and validated by the National Health Research Institute, contains medical data from 98.29% of the 23 million people in Taiwan [15]. Moreover, its diagnostic and procedural codes are based on the International Classification of Diseases, Ninth Revision, Clinical Modification (ICD-9-CM). The NHIRD contains information including patient characteristics, inpatient and outpatient visit records, prescriptions, disease states and management, and place of registration.

A PD patient is defined by a diagnosis of PD (ICD-9-CM code 332) and prescriptions of Levodopa in at least three office visits over 12 months. Prevalence is defined as the number of patients divided by the population in a specific year. The standardization of prevalence was based on the world standard population of the World Health Organization (WHO) in 2000 [16]. Here we studied the prevalence from 2009 to 2013.

### 2.2. Definition of the place of residence

We used the patient's address as the place of residence.

### 2.3. Heavy metal levels in the farm soils

There are three surveys on the soil contents of heavy metals in Taiwan. The data used in the study came from the most extensive one done from 1983 to 1986 [16–18]. The latter two surveys focused on areas where the contents of heavy metals were high. The mean of concentrations from all samples from the same township was used in the analysis. Taiwan was divided into 15 regions to include the hottest spots in one region. A hot spot is defined as a township where the majority of soil metal levels are higher than the average. The mean of every township's level represents the level for a region [16].

### 2.4. Self-correlation statistic based on Moran's *I*

The spatial autocorrelation analysis concerns the degree of the "Neighborhood Effect" [19]. Moran's Index (Moran's *I*) is a spatial autocorrelation method that estimates the size of the spatial association between all locations in a study area. The results by Moran's *I* range between  $-1$  and  $1$ . When the result is close to  $1$ , spatial clustering is present. On the other hand, when the result is negative, spatial clustering is absent. Moran's *I* is widely used in public health or environmental issues to verify whether the variables have spatial clustering effects [16,20,21].

### 2.5. Spatial regression model

Traditional regression models verify the causal relationship between two variables, not considering the spatial effects [22]. In contrast, the spatial regression model takes into account both the numerical and spatial relationships and incorporates the effects of spatial autocorrelation in a regression model [23]. In this paper, we used the spatial error model to perform spatial regression testing.

### 2.6. Statistical analysis procedure

We did a spatial autocorrelation analysis using PD prevalence and heavy metal levels. We then incorporated heavy metal levels in a spatial

regression model to study their relationship. Briefly, we used the spatial regression model to examine the independent variables (soil levels of heavy metals) effects on the dependent variable (PD prevalence). Moran's *I* was used to test the spatial autocorrelation of these variables before the spatial regression model. A variable must be significant in spatial autocorrelation to ensure its inclusion in the subsequent spatial regression model.

We used the R version 3.5.3 (The R Project for Statistical Computing, <https://www.r-project.org/>), and used the "ape" package for Moran's *I* test and the "spdep" package for the spatial regression test.

## 3. Results

### 3.1. The PD prevalence in taiwan

We divided Taiwan into 15 regions sorted geographically. The PD prevalence in each region is listed in Table 1. Because the PD prevalence is relatively stable, we used the mean PD prevalence from 2009 to 2013 for the following analysis. The prevalence ranged from 83 per one hundred thousand persons at Hualien to 213 at Yunlin. Fig. 1A shows the prevalence on a Taiwan map. The regions with high PD prevalence are located in central and southern Taiwan.

### 3.2. A strong correlation between soil as levels and the PD prevalence

We used the Local Moran's *I* to test the spatial autocorrelation of PD prevalence and heavy metal levels in farm soils. The spatial autocorrelation results (Table 2) shows PD prevalence in significant spatial correlation with As, Cu, Hg, Ni, Pb, and Zn. Subsequently, we used the spatial regression model to study the relationship between PD prevalence and the 6 heavy metals with significance. The spatial regression testing results (Table 3) show that only arsenic is associated with PD prevalence.

Arsenic stands out as the only metal associated with PD prevalence. Fig. 1B shows the soil As levels in Taiwan. Fig. 2 shows the scatter plot of PD prevalence verse soil As levels. There is a strong correlation ( $r$ -value 0.75).

## 4. Discussion

We found a strong correlation between PD prevalence and soil As levels. In other words, an increased PD prevalence was found where there were increased soil As levels.

Arsenic is ranked first among toxicants and poses a significant threat to human health [24]. More than 100 million people worldwide are exposed to excessive amounts of arsenic. Many of these people drink water from wells or live in regions where soil arsenic levels are high. While the World Health Organization and the Environmental Protection Agency regulate water sources of arsenic, there is an alarming lack of regulations on the other sources, e.g., farm soils or food [25].

Arsenic can accumulate in many parts of the brain [26] and induce toxicity through many mechanisms [27]. Despite many other heavy metals being reported to be involved in PD [28], we found only As was associated with PD prevalence in the study. These results suggest that As, compared to other heavy metals, poses a greater risk for PD. Another possibility is that As in the soils can accumulate in plants and readily be absorbed by humans.

Arsenic in drinking water has drawn the attention of many researchers and led to revisions of its regulations. By comparison, arsenic in farm soils has drawn little attention. In theory, humans can take up the metal in the soils via vegetables or crops growing in contaminated soils. This study highlights the serious risk of As in farm soils for humans.

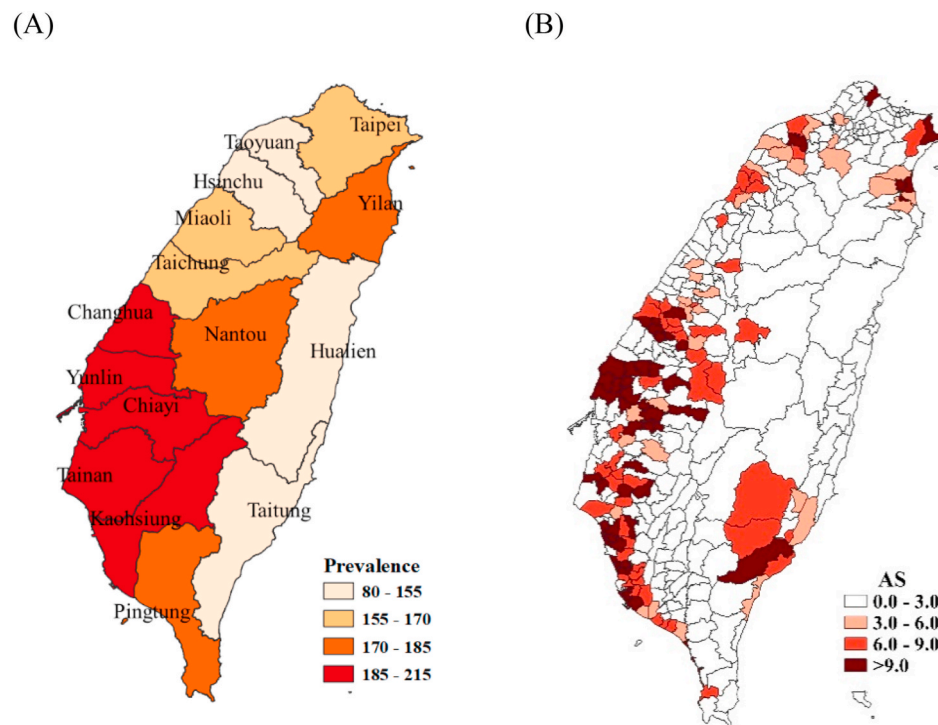
The information on soil metal concentrations is retrieved from national surveys by the government. There is a range of 30 years between collection of As data (1983–1986) and the PD prevalence (2009–2013). Metal contents in the soils are very stable unless there is continuous

**Table 1**

Age-standardized prevalence of PD and As levels in each area of Taiwan from 2009 to 2013.

| Area            | City      | Prevalence |      |      |      |      | Mean | As levels |
|-----------------|-----------|------------|------|------|------|------|------|-----------|
|                 |           | 2009       | 2010 | 2011 | 2012 | 2013 |      |           |
| Northern Taiwan | Taipei    | 170        | 173  | 164  | 156  | 161  | 169  | 2.69      |
|                 | Taoyuan   | 150        | 140  | 125  | 110  | 108  | 126  | 4.05      |
|                 | Hsinchu   | 146        | 170  | 170  | 143  | 150  | 147  | 3.68      |
|                 | Miaoli    | 159        | 166  | 160  | 166  | 169  | 165  | 3.47      |
| Central Taiwan  | Taichung  | 163        | 182  | 173  | 159  | 162  | 165  | 2.89      |
|                 | Nantou    | 166        | 194  | 168  | 176  | 172  | 175  | 6.24      |
|                 | Changhua  | 160        | 166  | 171  | 176  | 174  | 170  | 8.56      |
|                 | Yunlin    | 209        | 199  | 213  | 223  | 223  | 213  | 11.47     |
| Southern Taiwan | Chiayi    | 207        | 199  | 189  | 193  | 172  | 205  | 10.58     |
|                 | Tainan    | 203        | 207  | 185  | 191  | 211  | 199  | 8.86      |
|                 | Kaohsiung | 179        | 187  | 185  | 195  | 199  | 189  | 8.19      |
|                 | Pingtung  | 172        | 169  | 166  | 195  | 192  | 179  | 6.34      |
| Eastern Taiwan  | Yilan     | 173        | 199  | 187  | 180  | 209  | 183  | 7.57      |
|                 | Hualien   | 106        | 118  | 69   | 54   | 71   | 83   | 2.18      |
|                 | Taitung   | 153        | 117  | 97   | 96   | 100  | 113  | 4.86      |

Note: The unit for prevalence is per one hundred thousand persons.

**Fig. 1.** (A) The PD standardized prevalence in Taiwan; (B) Geographic distribution and farm soil arsenic levels at each township. The unit for soil heavy metal is mg/kg, and the unit for prevalence is per one hundred thousand persons.**Table 2**Local Moran's *I* for PD prevalence and heavy metals.

| Variables     | Moran's <i>I</i> | <i>p</i> -value |
|---------------|------------------|-----------------|
| PD Prevalence | 0.64             | <0.001*         |
| As            | 0.49             | 0.002*          |
| Cd            | 0.19             | 0.147           |
| Cr            | 0.18             | 0.124           |
| Cu            | 0.22             | 0.027*          |
| Hg            | 0.71             | <0.001*         |
| Ni            | 0.25             | 0.020*          |
| Pb            | 0.24             | 0.007*          |
| Zn            | 0.22             | 0.046*          |

\*\*\* denotes significance: *p*-value less than 0.05.

PD: Parkinson's disease.

**Table 3**

Spatial Regression results of Parkinson's disease prevalence and heavy metals.

|           | Estimate | S.E.  | z-value | <i>p</i> -value |
|-----------|----------|-------|---------|-----------------|
| Intercept | 113.28   | 24.8  | 4.56    | <0.001          |
| As        | 4.58     | 2.03  | 2.26    | 0.02*           |
| Cu        | −0.01    | 0.96  | −0.01   | 0.99            |
| Hg        | 40.25    | 29.17 | 1.38    | 0.17            |
| Ni        | −0.69    | 1.28  | −0.54   | 0.59            |
| Pb        | 0.23     | 0.34  | 0.69    | 0.49            |
| Zn        | −0.15    | 0.68  | −0.22   | 0.83            |

\*\*\* denotes significance: *p*-value less than 0.05.

input from human activities. As detailed in the introduction, arsenic in the soils came from well water, and irrigation with the well water was started in 1920. If there are no human efforts to remove metal from the

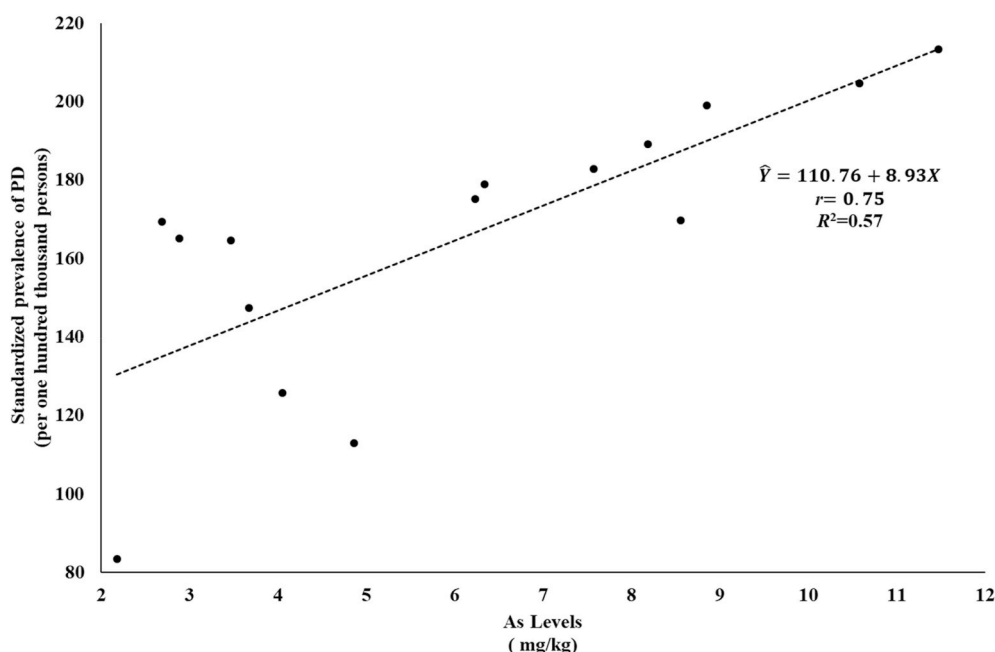


Fig. 2. The scatter plot of PD standardized prevalence vs. As levels.

soils, the metals will remain there. Arsenic in farm soils can pose as a real risk for PD. We can expect that PD prevalence and incidence will remain high in these areas unless there are human efforts to limit agriculture activities or to remove metals from the soils.

Several studies have assessed environmental factors for PD [29]. Increased risk of PD has been associated with exposure to pesticides, consumption of dairy products, history of melanoma, and traumatic brain injury. The information, unfortunately, helps little in the prevention of PD for ordinary people. We still take dairy products, enjoy sunlight, avoid head trauma, and use no pesticides. On the other hand, a reduced risk has been reported in association with smoking, caffeine consumption, higher serum urate concentrations, physical activity, and the use of ibuprofen and other common medications. Except for physical activities, it is unlikely that these approaches will be recommended to reduce PD risk. The key is to identify risks that are manageable and have a great neuroprotective effect before the onset of motor symptoms. Arsenic exposure can be a real significant risk, which is both manageable and neuroprotective when removed or reduced in exposure levels.

An intriguing hypothesis states that environmental factors can spread to the Central Nervous System (CNS) via the enteric nervous system (ENS) [30]. The gut-brain axis can play a significant role in PD pathogenesis. To our interest, As can significantly alter the gut microbiota in laboratory animals [31], and dysfunctional gut microbiota may predispose individuals to an increased PD risk. Does As act directly on brain neurons? Or act on the enteric nervous system, via retrograde pathways to cause structural changes in the brain? More studies are needed to answer these questions. Before that, a human study to correlate soil As levels and blood As levels is required, similar to the one we did in 2011 [32].

For an epidemiologic study, diagnostic criteria are of importance. Unfortunately, it is beyond the authors' ability to study the diagnostic criteria applied in each case or to validate the diagnosis. We acknowledge that the lack of examination of the patients is a shortcoming of the study. In Taiwan, patients with movement disorders are referred to neurologists who are authorized to prescribe Levodopa. Otherwise, the expenditure will not be reimbursed by insurance. Our selection criteria that a PD case must have at least three visits in one year means they are followed up by neurologists regularly and continuously. In the future, a study correlating blood and soil As levels, blood As levels, and hair As

contents in PD patients and aged-matched counterparts is envisioned.

## 5. Conclusion

We found that high soil arsenic correlated with high PD prevalence in Taiwan. Farm soil metal levels should be regulated. Arsenic is a risk that can be greatly minimized through human efforts to prevent PD before the onset of motor symptoms.

## Statement of ethics

This study was approved by the Office of Human Subject Protection, Changhua Christian Hospital (IRB no. 141008).

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None.

## Declaration of competing interest

The authors have no conflicts of interest to declare.

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